

Using Large Language Models to Support ERP Requirement Classification: An Empirical Study – Supplementary Materials

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1 Additional results for RQ1: Suitability of LLM configurations

Table 1: Performance summary measures for particular LLM configurations in dataset A. Values in braces indicate SD. Rows sorted by descending MCC.

Platform	Model	MCC	F1	Precision	Recall	Accuracy
Claude	Opus 4.6	0.630 (0.051)	0.811 (0.025)	0.829 (0.030)	0.802 (0.027)	0.834 (0.022)
Claude	Sonnet 4.6 Thinking	0.620 (0.038)	0.803 (0.019)	0.832 (0.022)	0.789 (0.019)	0.831 (0.016)
Claude	Sonnet 4.6	0.618 (0.057)	0.804 (0.025)	0.826 (0.039)	0.793 (0.022)	0.829 (0.025)
Claude	Opus 4.6 Thinking	0.605 (0.040)	0.801 (0.020)	0.808 (0.023)	0.797 (0.020)	0.822 (0.019)
MS 365 Copilot	GPT 5.4	0.578 (0.056)	0.772 (0.035)	0.823 (0.050)	0.762 (0.040)	0.808 (0.026)
Gemini	Gemini 3.1 Pro	0.573 (0.044)	0.761 (0.026)	0.836 (0.034)	0.745 (0.031)	0.807 (0.016)
Perplexity	GPT 5.4 Thinking	0.563 (0.081)	0.777 (0.046)	0.732 (0.114)	0.780 (0.035)	0.792 (0.050)
ChatGPT	GPT 5.4 Thinking	0.556 (0.036)	0.769 (0.022)	0.769 (0.021)	0.787 (0.015)	0.780 (0.025)
MiniMax	MiniMax-M2.7-Max	0.493 (0.064)	0.737 (0.037)	0.768 (0.033)	0.728 (0.036)	0.777 (0.027)
Z	GLM-5	0.446 (0.096)	0.668 (0.078)	0.810 (0.035)	0.666 (0.060)	0.757 (0.040)
MiniMax	MiniMax-M2.7-Air	0.438 (0.080)	0.677 (0.069)	0.786 (0.045)	0.674 (0.060)	0.754 (0.034)
Gemini	Gemini 3.1 Thinking	0.432 (0.061)	0.630 (0.052)	0.848 (0.017)	0.636 (0.036)	0.744 (0.025)
Z	GLM-5-Deep	0.413 (0.061)	0.627 (0.050)	0.827 (0.041)	0.633 (0.035)	0.739 (0.023)
DeepSeek	DeepSeek-V3.2	0.369 (0.079)	0.639 (0.034)	0.750 (0.058)	0.637 (0.029)	0.730 (0.027)
Claude	Haiku 4.5	0.344 (0.091)	0.637 (0.071)	0.718 (0.043)	0.641 (0.061)	0.719 (0.033)
Claude	Haiku 4.5 Thinking	0.282 (0.106)	0.586 (0.087)	0.713 (0.060)	0.605 (0.064)	0.701 (0.032)
Grok	Grok-4.20-Expert	0.263 (0.175)	0.548 (0.099)	0.721 (0.148)	0.582 (0.070)	0.700 (0.048)
Perplexity	Gemini 3.1 Pro Thinking	0.179 (0.036)	0.530 (0.053)	0.637 (0.101)	0.557 (0.025)	0.665 (0.013)

Table 2: Performance summary measures for particular LLM configurations in dataset B. Values in braces indicate SD. Rows sorted by descending MCC.

Platform	Model	MCC	F1	Precision	Recall	Accuracy
Perplexity	GPT 5.4 Thinking	0.589 (0.048)	0.796 (0.025)	0.700 (0.133)	0.786 (0.023)	0.816 (0.021)
ChatGPT	GPT 5.4 Thinking	0.558 (0.059)	0.777 (0.031)	0.763 (0.090)	0.770 (0.034)	0.805 (0.025)
Claude	Sonnet 4.6 Thinking	0.524 (0.065)	0.737 (0.038)	0.812 (0.032)	0.721 (0.035)	0.793 (0.026)
Claude	Opus 4.6 Thinking	0.518 (0.085)	0.749 (0.049)	0.780 (0.038)	0.740 (0.051)	0.790 (0.034)
Gemini	Gemini 3.1 Pro	0.515 (0.106)	0.702 (0.085)	0.853 (0.026)	0.693 (0.068)	0.785 (0.044)
Claude	Sonnet 4.6	0.514 (0.045)	0.738 (0.029)	0.797 (0.024)	0.723 (0.028)	0.790 (0.018)
Claude	Opus 4.6	0.486 (0.064)	0.731 (0.034)	0.770 (0.036)	0.719 (0.033)	0.778 (0.025)
MS 365 Copilot	GPT 5.4	0.470 (0.098)	0.725 (0.057)	0.751 (0.048)	0.722 (0.057)	0.765 (0.041)
Perplexity	Gemini 3.1 Pro Thinking	0.443 (0.158)	0.667 (0.113)	0.794 (0.109)	0.666 (0.080)	0.762 (0.054)
MiniMax	MiniMax-M2.7-Max	0.432 (0.163)	0.669 (0.128)	0.792 (0.051)	0.679 (0.101)	0.759 (0.056)
Z	GLM-5	0.388 (0.086)	0.616 (0.067)	0.812 (0.044)	0.625 (0.049)	0.737 (0.031)
MiniMax	MiniMax-M2.7-Air	0.353 (0.077)	0.658 (0.050)	0.705 (0.038)	0.654 (0.044)	0.727 (0.026)
Claude	Haiku 4.5	0.347 (0.106)	0.643 (0.060)	0.702 (0.112)	0.635 (0.066)	0.714 (0.068)
DeepSeek	DeepSeek-V3.2	0.324 (0.108)	0.599 (0.088)	0.722 (0.081)	0.613 (0.060)	0.721 (0.035)
Z	GLM-5-Deep	0.320 (0.096)	0.555 (0.083)	0.827 (0.037)	0.587 (0.050)	0.713 (0.030)
Claude	Haiku 4.5 Thinking	0.303 (0.117)	0.593 (0.072)	0.724 (0.078)	0.605 (0.045)	0.714 (0.031)
Gemini	Gemini 3.1 Thinking	0.265 (0.083)	0.519 (0.069)	0.805 (0.044)	0.564 (0.040)	0.698 (0.025)
Grok	Grok-4.20-Expert	0.214 (0.112)	0.520 (0.074)	0.716 (0.108)	0.560 (0.044)	0.688 (0.028)

2 Definitions of GLMMs used in RQ2.

Table 3: Definitions of GLMMs used in RQ2.

Model	Analysis group	Data scope	Fixed effects	Random intercepts
H0	Req. heterogeneity	Full	dataset	platform_model + run_id
Hreq	Req. heterogeneity	Full	dataset	platform_model + run_id + req_id
H0+act	Req. heterogeneity	Full	dataset + actual_class	platform_model + run_id
Hreq+act	Req. heterogeneity	Full	dataset + actual_class	platform_model + run_id + req_id
R0	Req. factors	Full	dataset	platform_model + run_id + req_id
Ract	Req. factors	Full	dataset + actual_class	platform_model + run_id + req_id
Rlen	Req. factors	Full	dataset + length_chars.z	platform_model + run_id + req_id
R0(c)	Req. factors	Common	dataset	platform_model + run_id + req_id
Rarea(c)	Req. factors	Common	dataset + area	platform_model + run_id + req_id
Ract(c)	Req. factors	Common	dataset + actual_class	platform_model + run_id + req_id
Ract+len	Req. factors	Full	dataset + actual_class + length_chars.z	platform_model + run_id + req_id
Ract+area(c)	Req. factors	Common	dataset + actual_class + area	platform_model + run_id + req_id
Ract*len	Req. factors	Full	dataset + actual_class × length_chars.z	platform_model + run_id + req_id

All models are binomial GLMMs with logit link.

Models were fitted using maximum likelihood with Laplace approximation via `lme4::glmer` in **R**.

The suffix '(c)' denotes models fitted on the common-area subset.

Data scope 'common' denotes a reduced dataset containing area categories that appear in both datasets A and B.

The variable `length_chars.z` denotes standardised requirement description length.

3 Additional results for RQ2:

Table 4: Detailed comparison of paired GLMMs for the RQ2 (Part 1).

Family	Model 1				Model 2				
	Abbr.	AIC	R_m^2	R_c^2	Abbr.	AIC	R_m^2	R_c^2	MOR
Baseline	H0	40457.734	0.000	—	Hreq	21929.172	0.001	0.735	17.081
Baseline	H0+act	29840.095	0.345	0.365	Hreq+act	21779.597	0.371	0.715	6.265
Standalone	R0	21929.172	0.001	0.735	Ract	21779.597	0.371	0.715	6.265
Standalone	R0	21929.172	0.001	0.735	Rlen	21921.555	0.035	0.735	15.975
Standalone	R0(c)	18728.572	0.006	0.757	Rarea(c)	18693.720	0.188	0.760	13.700
Adjusted	Ract	21779.597	0.371	0.715	Ract+len	21781.436	0.372	0.715	6.266
Adjusted	Ract(c)	18603.207	0.384	0.739	Ract+area(c)	18603.741	0.407	0.741	6.633
Interaction	Ract+len	21781.436	0.372	0.715	Ract*len	21771.041	0.392	0.715	5.888

R_m^2 and R_c^2 marginal and conditional Nakagawa R^2 for particular models.

MOR is the median odds ratio for model 2 computed from the requirement-level random intercept.

Table 5: Detailed comparison of paired GLMMs for the RQ2 (Part 2).

Family	Model 1	Model 2	LR χ^2	Δdf	p	p_{adj}	ΔAIC	ΔR_c^2	ΔR_m^2
Baseline	H0	Hreq	18 530.562	1	<0.001	<0.001	18528.562	—	0.001
Baseline	H0+act	Hreq+act	8062.498	1	<0.001	<0.001	8060.498	0.350	0.026
Standalone	R0	Ract	151.575	1	<0.001	<0.001	149.575	−0.020	0.370
Standalone	R0	Rlen	9.616	1	0.002	0.002	7.616	0.000	0.033
Standalone	R0(c)	Rarea(c)	46.852	6	<0.001	<0.001	34.852	0.003	0.182
Adjusted	Ract	Ract+len	0.160	1	0.689	0.689	−1.840	0.000	0.000
Adjusted	Ract(c)	Ract+area(c)	11.466	6	0.075	0.150	−0.534	0.002	0.023
Interaction	Ract+len	Ract*len	12.395	1	<0.001	<0.001	10.395	−0.001	0.020

LR χ^2 and p -values come from likelihood-ratio tests of compared models.

Raw p -values were adjusted within each family using the Holm method.

$\Delta AIC = AIC_{Model1} - AIC_{Model2}$, so positive values favour Model 2.

ΔR_m^2 and ΔR_c^2 denote changes in ΔR_m^2 and ΔR_c^2 from Model 1 to Model 2.